ADAPTIVE CLASSROOM ENERGY MANAGEMENT

MINI PROJECT

TEAM GUIDE

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INTRODUCTION

- Classroom appliances like fans, lights are manually controlled, causing energy wastage

- An automated system using **PIR**, **temperature**, and **light** sensors can intelligently control appliances by detecting student presence,ambient temperature,and natural light availability, ensuring efficient energy usage

PROBLEM STATEMENT

Manual control of classroom lights and fans without considering occupancy, light intensity, and temperature causes unnecessary energy wastage. An automated system is required to control appliances efficiently based on room occupancy and environmental conditions

AIM

To design an automatic classroom energy management system that controls lights and fans based on occupancy, ambient light, and temperature to reduce energy wastage

OBJECTIVES

- To automatically control lights and fans based on real-time environmental conditions.
- To sense classroom conditions such as light intensity, temperature, and occupancy, the system sends signals to the microcontroller, which automatically switches the lights and fan ON or OFF to reduce unnecessary power consumption and maintain basic comfort
- To demonstrate a simple system where a single LED light (9– 12 W) and a fan (35–60 W) connected through an extension board which operate automatically only when needed.
- To display classroom conditions such as temperature (in Celsius), light intensity, motion detection, and the status of the fan and lights (ON/OFF), an **OLED** display is used.
- To provide two manual switch option for controlling the fan and lights when required.

LITERATURE SURVEY

PAPER TITLE/SOURCE	RESEARCH CONTRIBUTION	RESEARCH GAPS	REFERENCE LINK
Implementation of Automated Smart Classroom Using Sensors (IJIAREC, 2018)	Automates fan, lights, projector using sensors • Presence-based appliance control • Fan speed via temperature sensor • Arduino-based real-time system	• No measured energy-saving analysis • No PIR sensor for accurate detection • Limited classroom coverage • Not scalable for large rooms	https://ijiarec.com/ijiarec/article/view/445

PAPER TITLE/SOURCE	RESEARCH CONTRIBUTION	RESEARCH GAPS	REFERENCE LINK
Energy Saving System Using a PIR Sensor (IJRASET, 2023)	• PIR-based human detection • Automatic control of lights and fans • Microcontroller + relay switching • Reduces manual power usage	• Cannot count number of students • Limited PIR coverage • Fixed delay response • No ambient light-based control	https://www.ijraset.com/research-paper/energy-saving-system-using-a-pir-sensor

COMPONENTS

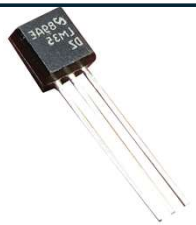
1.PIR SENSOR (HC SR-501)

The PIR sensor detects human motion or occupancy in a room. It operates at 5–12V and outputs a HIGH signal when motion is detected



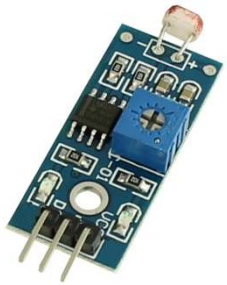
2.TEMPERATURE SENSOR (LM 35)

The LM35 is a precision temperature sensor that operates at 4–30V and can be used for temperature monitoring



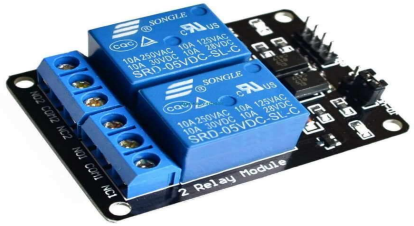
3.LDR MODULE (GL 5528)

The LDR module detects ambient light intensity using an LM393 comparator for adjustable sensitivity. It provides digital and analog outputs, enabling automation of devices like streetlights based on brightness levels.



4.RELAY MODULE

A relay module is an electrically operated switch that allows low-voltage signals to control high-voltage devices. It works with a 5V control signal and is used to safely turn on or off devices like fans and lights.



5.OLED DISPLAY

The OLED display is a low-power module with a resolution of 128×64 pixels. It operates at 3.3–5V and communicates using the I2C protocol. It is commonly used to show sensor readings and system status



6.ESP32D MICROCONTROLLER

The ESP32 is a powerful microcontroller with built-in Wi-Fi and Bluetooth for IoT applications. It operates at 3.0–3.6V and features a 240 MHz dual-core processor. It is widely used for control, monitoring, and communication in smart systems.



7.5 A Fuse

A 5A fuse is a protective device that safeguards circuits against overcurrent by breaking the circuit if the current exceeds 5 amperes. It ensures the safety of connected components and prevents potential damage or hazards.



8.Switch

A switch is a manual device used to open or close a circuit, allowing control of electrical devices like lights and fans. It provides a simple way to turn the system ON or OFF safely.



SYSTEM DESIGN

- The system consists of **input sensors (PIR HC-SR501, LM35, and LDR GL5528)** connected to the **ESP32 microcontroller**
- Sensor data is processed by the **ESP32**, which makes control decisions based on classroom conditions
- The **relay module** acts as an interface between the ESP32 and electrical appliances like **lights and fan**
- An **OLED display** is used to show real-time parameters such as temperature,light intensity (ADC value), motion status, and appliance status
- Two **manual switch** is provided to allow users to control the appliances when required

ESP32 Development Board

- Operating voltage = 3.3 V
- Used as the main controller
- Reads sensor data through GPIO and ADC pins
- Controls relay module and OLED display using digital signals and I2C communication

PIR Sensor (HC-SR501)

- Operating voltage = 5 V
- Output type = Digital
- Output condition:
- Motion detected → HIGH
- No motion → LOW
- Used to activate system operation based on human motion detection
- Detection distance: **3 to 7 meters**
- Detection angle: **~110° to 140°**

LDR Sensor (GL5528)

- Works using voltage divider principle:
  Vout=Vin×R/(R+RLDR)
- Vin = 3.3 V
- Output read using ADC pin of ESP32
- Threshold selected experimentally for automatic light control
- Condition:
  Low light → Light ON
  High light → Light OFF
- The analog voltage is converted into digital value using ESP32 ADC(12-bit resolution: 0 – 4095)
- Threshold is selected experimentally based on room condition
- ADC threshold = 1600
- ADC resolution = 12-bit = 4095
- Using ADC formula:

$$V_{out} = \frac{ADC}{4095} \times 3.3$$

- Voltage divider equation:

$$R_{LDR} = R\left(\frac{V_{in}}{V_{out}} - 1\right)$$

Where:

Fixed resistor $R = 10k\Omega$

$$V_{in} = 3.3V$$
$$V_{out} = 1.29V$$

$$Lux = \left(\frac{500}{R_{\{LDR\}}}\right)^{\{1.4\}}$$

Substitute:

$$R_{LDR} = 10k\left(\frac{3.3}{1.29} - 1\right)$$
$$R_{LDR} = 10k(2.56 - 1)$$
$$R_{LDR} \approx 15.6k\Omega$$

Where:

Lux= illumination in lux

R_{LDR} =LDR resistance in kΩ

If the resistance is:

$$R_{LDR} = 15.6k\Omega$$

Then,

$$Lux = \left(\frac{500}{15.6}\right)^{1.4}$$

$$Lux = (32.05)^{1.4}$$

$$Lux \approx 128\ lux$$

Considering practical sensor variation and experimental calibration, it is approximated as:

$$Lux \approx 130–200\ lux$$

for the selected threshold value.

Substituting values:

- For this system:
- Threshold ≈ 1600 (≈ 200 lux)

LM35 Temperature Sensor

- Temperature range: **−55°C to +150°C**
- Supply voltage: **4V – 30V**
- Output type: **Analog voltage**

ADC resolution = 12-bit (0–4095)

- Output relation:

$$V_{out} = 10mV \times T$$

- Output relation in volts:

Vout=0.01×T

Where:

T= temperature in °C

For threshold temperature:

$$T = 24^{\circ}C$$
$$V_{out} = 10mV \times 24$$
$$V_{out} = 240mV = 0.24V$$

ESP32 ADC formula:

$$ADC = \frac{V_{out}}{3.3} \times 4095$$

Substitute $V_{out} = 0.3V$:

$$ADC = \frac{0.24}{3.3} \times 4095$$

$$ADC \approx 298$$

Condition:

Temperature ≥ 24°C → Fan ON

Fan

- Supply voltage = 230V AC
- Power rating ≈ 60 W
- Controlled automatically using relay based on temperature and motion detection

Light

- Supply voltage = 230V AC
- Power rating ≈ 9 W LED bulb
- Controlled automatically based on motion detection and light intensity

Relay Module (5V)

Used to control 230V AC appliances

- Load calculation:
  - Fan power ≈ 60 W
    - $I=V/P=60/230\approx0.26$ A
  - Light power ≈ 9 W
    - $I = V/P=9/230= 0.04$A
- Total current:
  - 0.26 + 0.04 = 0.30A
- Relay rating:
  - 5A / 230V AC

Fuse (5A)

Fuse selection rule:

Fuse\ rating ≈ 2 × Load\ current

Load current ≈ 0.30A

Selected fuse = 5A

Provides safety against overcurrent and short circuit

Switch

- Used for manual control and safety isolation
- Allows user to operate system manually if required

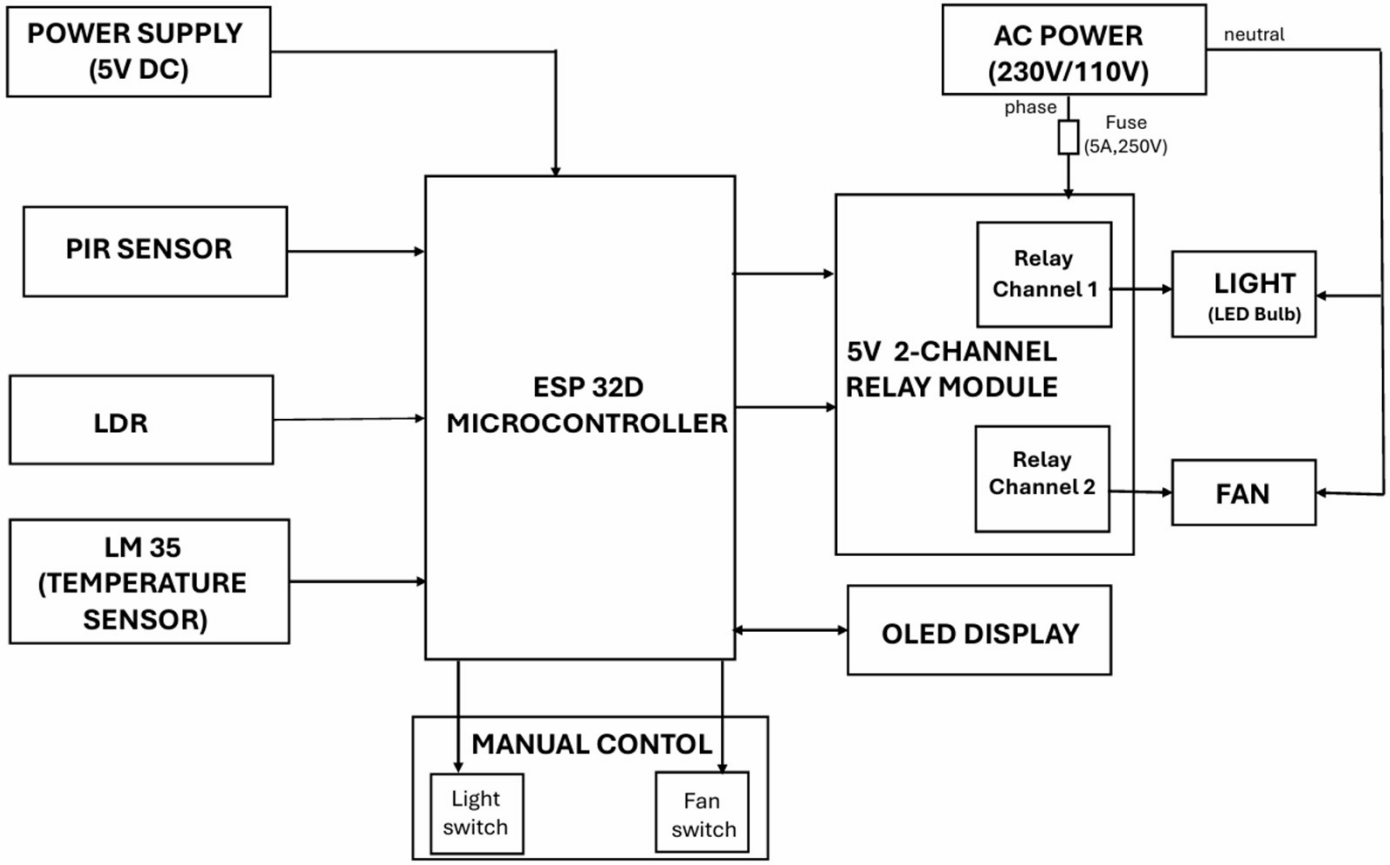
OLED Display

- Operating voltage = 3.3V–5V
- Communication protocol = I2C
- Pins used:

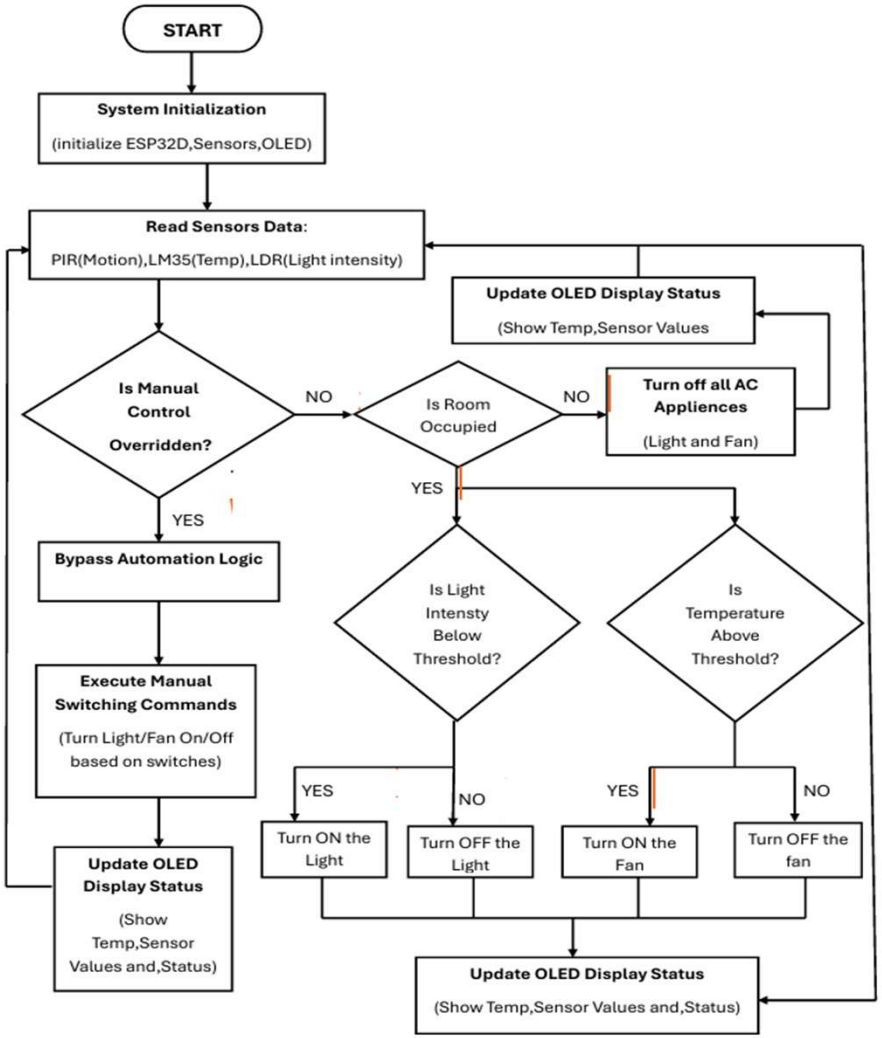
  SDA

  SCL
- Displays system status like motion detection and temperature value

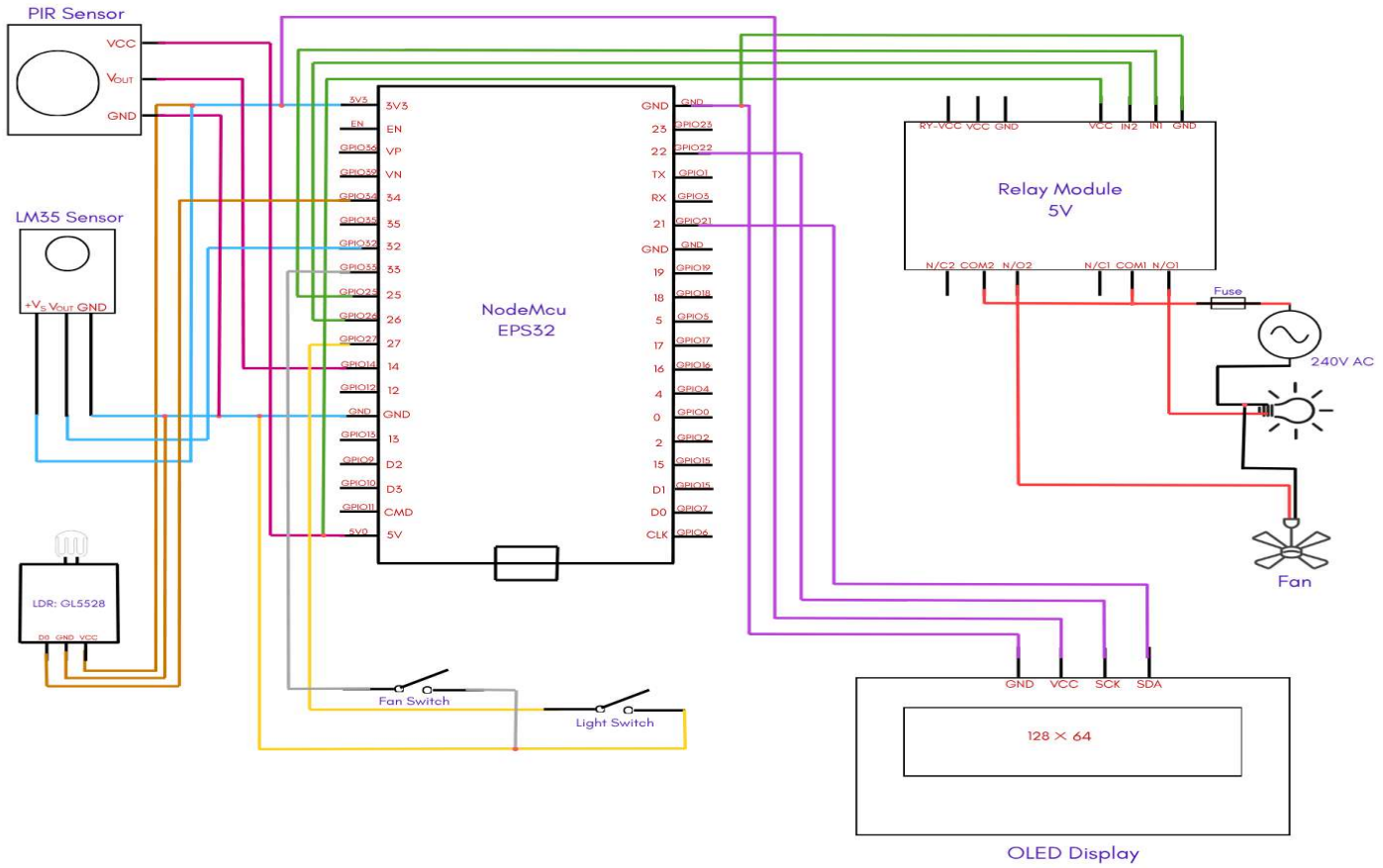
BLOCK DIAGRAM



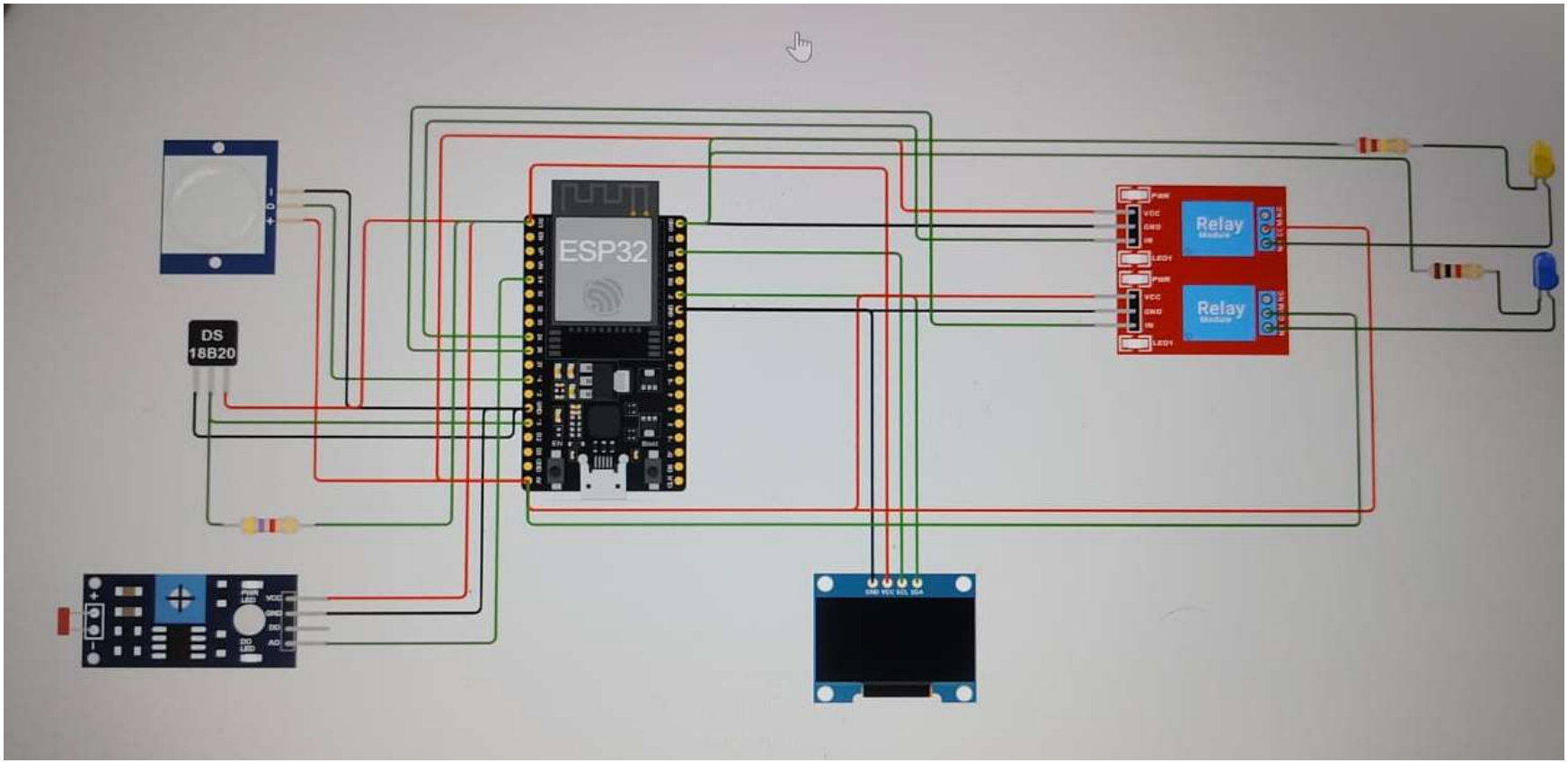
FLOWCHART


START
System Initialization
(initialize ESP32D,Sensors,OLED)
Read Sensors Data:
PIR(Motion),LM35(Temp),LDR(Light intensity)
Update OLED Display Status
(Show Temp,Sensor Values
Is Manual Control Overridden?
NO
Is Room Occupied
NO
Turn off all AC Appliences
(Light and Fan)
YES
YES
Bypass Automation Logic
Is Light Intensty Below Threshold?
Is Temperature Above Threshold?
Execute Manual Switching Commands
(Turn Light/Fan On/Off based on switches)
YES
NO
YES
NO
Turn ON the Light
Turn OFF the Light
Turn ON the Fan
Turn OFF the fan
Update OLED Display Status
(Show Temp,Sensor Values and,Status)
Update OLED Display Status
(Show Temp,Sensor Values and,Status)

CIRCUIT DIAGRAM



SIMULATION



:-fig: SIMULATION IN WOKWI

PRACTICAL IMPLEMENTATION

- The system is implemented as a **prototype model using ESP32 microcontroller** as the main controller.
- The **PIR sensor (HC-SR501)** is connected to detect classroom occupancy (motion).
- The **LM35 temperature sensor** is interfaced with ESP32 to measure room temperature.
- The **LDR (GL5528)** is used to sense classroom light intensity in ADC-converted mode.
- An **OLED display** is connected to show temperature, light intensity, motion status, and appliance ON/OFF status.

- A **relay module** interfaces the ESP32 with **AC appliances such as a fan and light** for demonstration
- An **AC power supply with fuse protection** is used to ensure safe and reliable operation.
- **Manual switches** are provided for user control during testing and operation.
- All components are assembled using a **breadboard prototype setup** for practical demonstration.

CONTROL LOGIC

HC-SR501 PIR SENSOR

- Detect occupancy based on infrared rays

LM35 TEMPERSENSORATURE

- No Occupancy: Fans OFF
- Occupancy Detected & Temperature < 24 °C: Fans OFF
- Occupancy Detected & Temperature => 24 °C: Fans ON

LDR GL5528 SENSOR

- No occupancy : Lights OFF
- Occupancy detected + sufficient daylight(<1600) : Lights OFF
- Occupancy detected + low light(>=1600): Lights ON

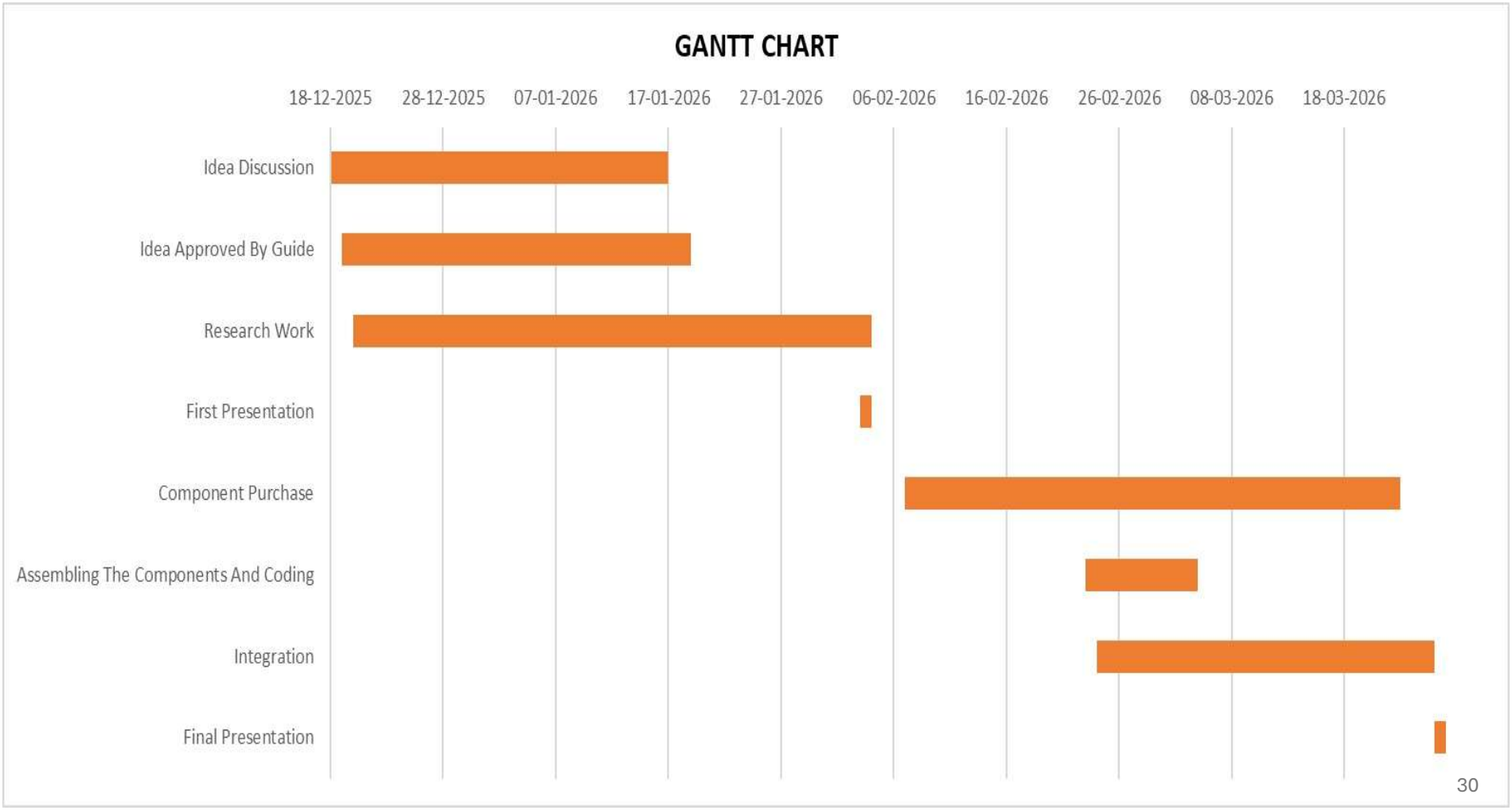
WORKING / METHODOLOGY

- **PIR Sensor:** Detects classroom occupancy via infrared; sends signal to ESP32.
- **Temperature Control (LM35):** Fan ON if occupancy detected and temperature ≥ 24 °C; otherwise OFF.
- **Light Control (LDR GL5528):** Lights ON if occupancy detected and ambient light is low; otherwise OFF.
- **ESP32 Controller:** Processes sensor data and switches fan and lights through relays
- **Display & Monitoring:** OLED shows occupancy, temperature, and light status.
- **Safety & Manual Control:** 5A fuse protects circuit; manual switch allows override.

TECHNICAL PARAMETERS AND SPECIFICATIONS

Component	Specification
Microcontroller(ESP32D)	**3.3 V logic**, input supply **5 V via regulator/USB**
Light Sensor (LDR GL-5528**)**	Operating voltage: **3.3–5 V**, Current: **~0.1–1 mA**
OLED Display	**3.3 V DC**, Current: **10–30 mA**
PIR Sensor (HC-SR501)	Supply: **5 V DC**, Current: **~50–65 µA , up to ~10 mA active**
Temperature Sensor(LM35)	Supply: **4–30 V DC**, Range: **0°C to 100°C**, Accuracy: **±0.5°C**, Output: **10 mV/°C**
Relay Module	Coil: **5 V DC**, Contact rating: **250 V AC / 10 A**
Fan Load	**230 V AC**, Power: **35–60 W**, Current: **0.15–0.26 A**
Light	**230 V AC**, Power: **9–12 W**, Current: **0.04–0.06 A**
Power Supply	Input: **230 V AC**, Output: **5 V DC (regulated)**

GANTT CHART


GANTT CHART
18-12-2025
28-12-2025
07-01-2026
17-01-2026
27-01-2026
06-02-2026
16-02-2026
26-02-2026
08-03-2026
18-03-2026
Idea Discussion
Idea Approved By Guide
Research Work
First Presentation
Component Purchase
Assembling The Components And Coding
Integration
Final Presentation
30

WORK DISTRIBUTION

PHASE	MEMBER	WORK ASSIGNED
PHASE 1: IDEA & PLANNING		
	Athul Krishna K H	Identified classroom energy wastage and proposed an automation solution.
	Fathimathu Saja K C	Prepared basic system block diagram.
	Salman Fariz T M	Defined objectives for automatic lighting and fan control.
	Shifanath T S	Selected components (ESP32, LDR, PIR, Relays).
PHASE 2: STUDY & SPECIFICATIONS		
	Athul Krishna K H	Studied ESP32 features and working.
	Fathimathu Saja K C	Analysed LDR sensor characteristics and implemented threshold setting techniques.
	Salman Fariz T M	Studied the PIR sensor and motion detection range.
	Shifanath T S	Examined relay module operation and analysed power requirements.

PHASE 3: DOCUMENTATION		
	Athul Krishna K H	Prepared the introduction and detailed problem statement.
	Fathimathu Saja K C	Documented hardware specification and component details.
	Salman Fariz T M	Developed the system working principle and flowchart design.
	Shifanath T S	Designed the circuit diagram and completed report formatting.
PHASE 4: COMPONENT PURCHASE PLANNING		
	Athul Krishna K H	Listed all required electronic and hardware components.
	Fathimathu Saja K C	Estimated total project cost and prepared budget analysis.
	Salman Fariz T M	Verified component ratings and compatibility.
	Shifanath T S	Planned wiring layout and ensured safety material selection.

PHASE 5: IMPLEMENTATION PLANNING		
	Athul Krishna K H	Planned coding framework for system implementation.
	Fathimathu Saja K C	Planned light control logic using an LDR sensor.
	Salman Fariz T M	Planned fan control mechanism using a PIR sensor.
	Shifanath T S	Planned system testing, integration and performance verification.

COST ESTIMATION

SI.NO	COMPONENTS	COST	SI.NO	COMPONENTS	COST
1	ESP32D	381	11	JUMPER WIRES(3 SET)	45
2	PIR SENSOR	60	12	DATA CABLE	150
3	TEMPERATURE SENSOR	40	13	BREADBOARD	130
4	LDR SENSOR	60	14	5 A FUSE	30
5	TWO CHANNEL 5V RELAY MODULE	80	15	THREE PIN PLUG	100
6	OLED DISPLAY	280	16	SOCKET PLUG	60
7	HLK-LD2410	590	17	SWITCH BOARD	150
8	LIGHT	120	18	CONNECTING WIRES	60
9	FAN	2100	19	SWITCHES	40
10	EXTENSION BOARD	250			

TOTAL COST: 4726/-

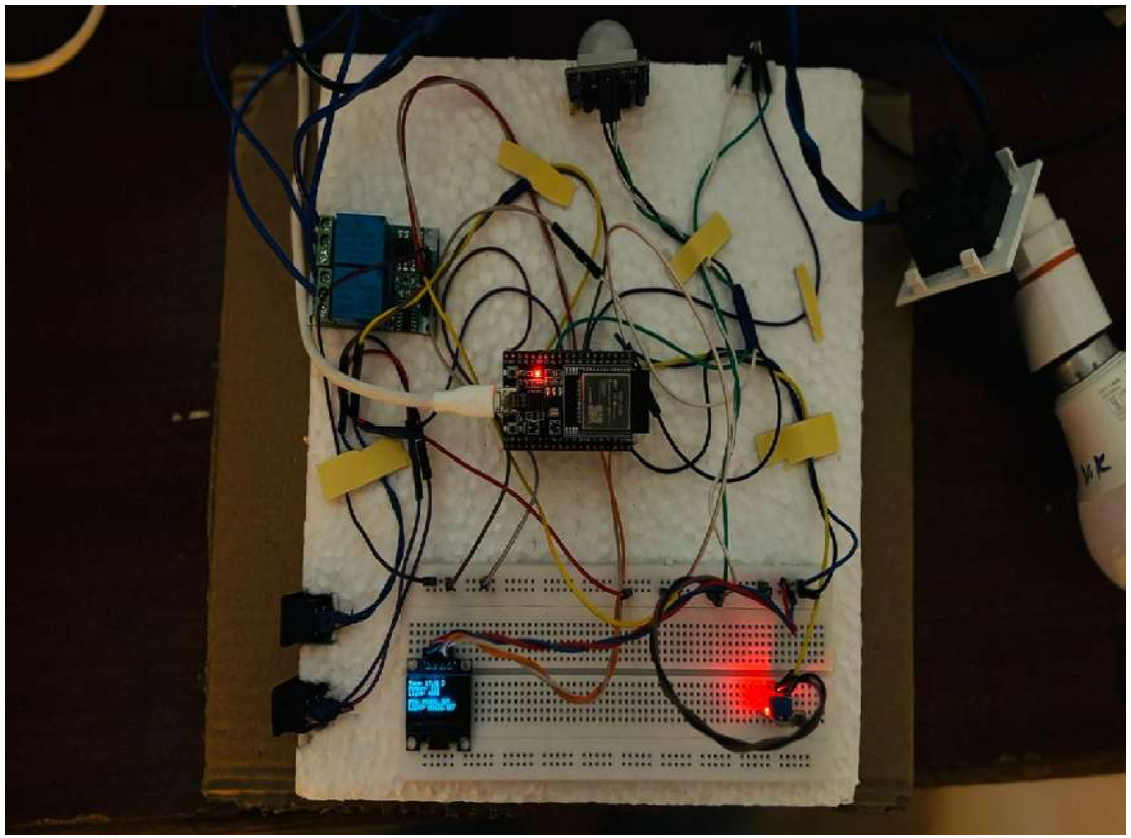
RESULT

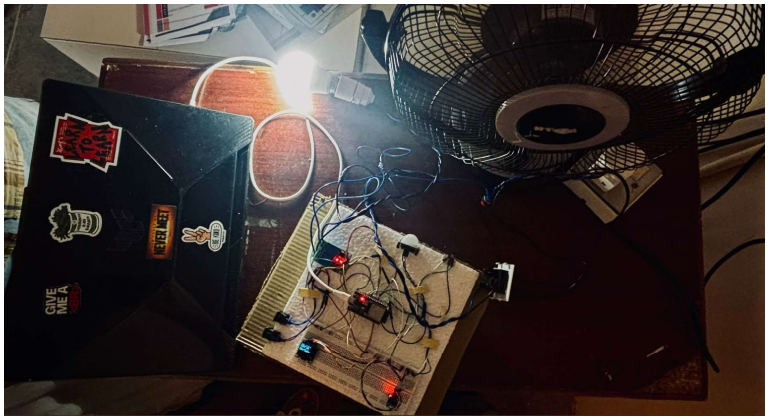
- The Smart Classroom Energy Management System was **successfully implemented using ESP32** with PIR sensor, LM35 temperature sensor, and LDR sensor
- The **PIR sensor detected human motion** and was used to determine classroom occupancy status.
- The **LM35 sensor monitored temperature** and automatically switched the **fan ON when temperature ≥ 22 °C** during motion detection.
- The **LDR sensor controlled the lights** based on surrounding light conditions.
- The **relay module successfully operated the AC fan and light** according to sensor inputs.
- The **OLED display showed real-time status** of motion detection, temperature, and appliance operation
- The system helped **reduce unnecessary power consumption** when the classroom was empty.

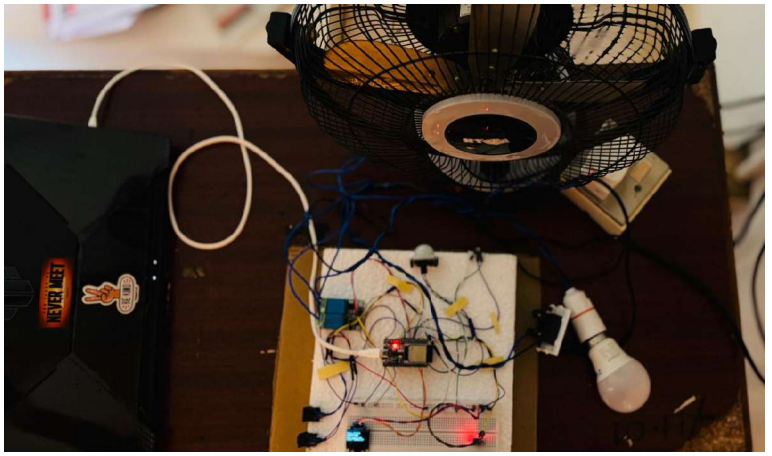
FUTURESCOPE

- Integration of IoT for remote monitoring and control
- Mobile app or web-based classroom management
- Suitable for smart classroom and smart campus systems
- Use of AI and machine learning for intelligent automation
- Automatic optimization based on usage patterns
- Improved energy efficiency using smart decision making
- Addition of humidity, air quality, and CO_2 sensors
- Enhanced classroom comfort and environmental monitoring
- Expansion to control projectors, ACs, and ventilation systems
- Cloud-based data storage and energy analysis
- Power usage monitoring for better management
- Use of solid-state relays for faster and reliable switching
- Improved scalability and system performance

HARDWARE







CONCLUSION

The Smart Classroom Energy Management System successfully demonstrated automatic control of classroom lights and fan based on occupancy,temperature, and ambient light conditions using the ESP32, thereby improving energy efficiency and reducing unnecessary power consumption.

CODE

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

// -------- PIN DEFINITIONS --------
#define PIR_PIN 14
#define LDR_PIN 34
#define LM35_PIN 32

#define RELAY_FAN 26
#define RELAY_LIGHT 25

#define FAN_SWITCH 33
#define LIGHT_SWITCH 27

// -------- OLED --------
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);
```

```
// -------- TIMING --------
unsigned long lastUpdate = 0;
const int interval = 200;

// -------- MOTION FILTER --------
bool motionState = false;
unsigned long lastMotionTime = 0;
unsigned long lastHighTime = 0;

const unsigned long motionHoldTime = 5000;
const unsigned long lowConfirmTime = 500;

// -------- MODES --------
int fanMode = 0;   // 0=AUTO, 1=ON, 2=OFF
int lightMode = 0; // 0=AUTO, 1=ON, 2=OFF


bool lastFanSwitch = HIGH;
bool lastLightSwitch = HIGH;
```

```
// -------- SETUP --------
void setup() {

  Serial.begin(115200);
  pinMode(PIR_PIN, INPUT);
  pinMode(LDR_PIN, INPUT);
  pinMode(LM35_PIN, INPUT);
  pinMode(RELAY_FAN, OUTPUT);
  pinMode(RELAY_LIGHT, OUTPUT);
  pinMode(FAN_SWITCH, INPUT_PULLUP);
  pinMode(LIGHT_SWITCH, INPUT_PULLUP);
  digitalWrite(RELAY_FAN, HIGH);
  digitalWrite(RELAY_LIGHT, HIGH);

delay(3000);

  Wire.begin(21, 22);

  if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    while (true);
  }

  display.clearDisplay();
  display.setTextSize(1);
  display.setTextColor(WHITE);
}


// -------- LOOP --------
void loop() {


  if (millis() - lastUpdate >= interval) {
    lastUpdate = millis();
```

```
// -------- SWITCH HANDLING --------
  bool fanSw = digitalRead(FAN_SWITCH);
  bool lightSw = digitalRead(LIGHT_SWITCH);


  if (lastFanSwitch == HIGH && fanSw == LOW) {
    fanMode++;
    if (fanMode > 2) fanMode = 0;
    delay(200);
  }


  if (lastLightSwitch == HIGH && lightSw == LOW) {
    lightMode++;
    if (lightMode > 2) lightMode = 0;
    delay(200);
  }


  lastFanSwitch = fanSw;
  lastLightSwitch = lightSw;


  int rawMotion = digitalRead(PIR_PIN);


  // -------- MOTION FILTER --------
  if (rawMotion == HIGH) {
    lastMotionTime = millis();
    lastHighTime = millis();
  }
```

```
if (rawMotion == LOW) {
    if (millis() - lastHighTime > lowConfirmTime) {
      if (millis() - lastMotionTime > motionHoldTime) {
        motionState = false;
      }
    }
  }


  if (millis() - lastMotionTime <= motionHoldTime) {
    motionState = true;
  }


  int lightValue = analogRead(LDR_PIN);


  int adcValue = analogRead(LM35_PIN);
  float voltage = adcValue * (3.3 / 4095.0);
  float temp = voltage * 100.0;


  // -------- FAN CONTROL --------
  if (fanMode == 1) {
    digitalWrite(RELAY_FAN, LOW); // ON
  }
  else if (fanMode == 2) {
    digitalWrite(RELAY_FAN, HIGH); // OFF
  }
  else {
    if (motionState && temp > 24) {
      digitalWrite(RELAY_FAN, LOW);
    } else {
      digitalWrite(RELAY_FAN, HIGH);
    }
  }
```

```
// -------- LIGHT CONTROL --------
  if (lightMode == 1) {
    digitalWrite(RELAY_LIGHT, LOW); // ON
  }
  else if (lightMode == 2) {
    digitalWrite(RELAY_LIGHT, HIGH); // OFF
  }
  else {
    if (motionState && lightValue >= 2000) {
      digitalWrite(RELAY_LIGHT, LOW);
    } else {
      digitalWrite(RELAY_LIGHT, HIGH);
    }
  }


  // -------- OLED --------
  display.clearDisplay();


  display.setCursor(0, 0);
  display.print("Temp: ");
  display.print(temp);
  display.println(" C");


  display.print("Motion: ");
  display.println(motionState ? "YES" : "NO");


  display.print("Light: ");
  display.println(lightValue);


  display.println("");
```

```
// FAN STATUS
   if (fanMode == 1) display.println("FAN: MANUAL ON");
   else if (fanMode == 2) display.println("FAN: MANUAL
OFF");
   else display.println((motionState && temp > 24) ? "FAN:
AUTO ON" : "FAN: AUTO OFF");


   // LIGHT STATUS
   if (lightMode == 1) display.println("LIGHT: MANUAL ON");
   else if (lightMode == 2) display.println("LIGHT: MANUAL
OFF");
   else display.println((motionState && lightValue >= 2000) ?
"LIGHT: AUTO ON" : "LIGHT: AUTO OFF");


   display.display();


   // -------- SERIAL --------
   Serial.print("FanMode: "); Serial.print(fanMode);
   Serial.print(" | LightMode: "); Serial.print(lightMode);
   Serial.print(" | Temp: "); Serial.print(temp);
   Serial.print(" | Motion: "); Serial.println(motionState);
  }
}
```

THANK YOU